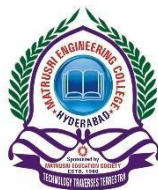


A Major Project Phase-II Report
On
"PENDULUM DRIVEN SPHERICAL ROBOT FOR
SURVEILLANCE OPERATION"

SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF DEGREE OF
BACHELOR OF ENGINEERING
IN
ELECTRONICS & COMMUNICATION ENGINEERING

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This is to certify that the Major Project Phase-II report entitled **“PENDULUM DRIVEN SPHERICAL ROBOT FOR SURVEILLANCE OPERATION”** being submitted by **Mr. K. BHARATH (1608-21-735-006), Ms. B.POOJITHA (1608-21-735-021) and Mr. M. RAHUL (1608-21-735-036)** partial fulfilment for the award of the Degree of Bachelor of Engineering in Electronics and Communication Engineering of the Osmania University, Hyderabad, during 2024-25, is a record of bonafide work carried out under our guidance and supervision.

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ABSTRACT

The project focuses on developing an innovative spherical robot designed to enhance surveillance in challenging and inaccessible environments. Unlike traditional humanoid, wheeled, or legged robots constrained by high costs, structural complexity, and terrain limitations, this robot employs a pendulum-based drive system for stable and efficient movement. Equipped with 360-degree cameras for real-time video streaming, it enables advanced monitoring and reconnaissance in unreachable areas. Powered by an ESP32-CAM controller, the robot integrates motors and feedback systems for precise operation. An Android application, connected via Bluetooth, facilitates intuitive remote control for movement and obstacle detection. Lightweight and durable, with an acrylic shell, this robot is ideal for military, search-and-rescue, and high-risk surveillance applications. Its modular design ensures easy maintenance, and future upgrades aim to incorporate full automation and advanced navigation techniques.

Keywords: Spherical robot, Pendulum-based drive system, Video streaming, ESP32-CAM controller.

CONTENTS

	Page No.
Certificate	i
Declaration	ii
Acknowledgment	iii
Abstract	iv
List of Figures	vii
List of Tables	viii
Chapter 1: INTRODUCTION	1
1.1 Introduction	1
1.2 Problem statement	2
1.3 Objective	2
1.4 Methodology	2
1.5 Motivation	3
1.6 Thesis Organization	3
Chapter 2: LITERATURE SURVEY	4
2.1 Introduction	4
2.2 Literature Reviews	4
Chapter 3: PROPOSED SYSTEM	12
3.1 Introduction	12
3.2 Block Diagram	12
3.3 Methodology	13
3.4 Flowchart	13
3.5 Applications	13
3.6 Advantages	14
3.7 Disadvantages	14
Chapter 4: HARDWARE IMPLEMENTATION	15
4.1 Introduction	15
4.2 Hardware Selection	15
4.2.1 ESP32 – CAM	15
4.2.2 LM298N Motor Driver	17
4.2.3 Shaft Geared Motor	18

4.2.4 Buck Convertor (LM2596)	19
4.2.5 MG996R Servo Motor	21
4.2.6 12V Battery (18650 Li-ion Battery)	22
4.2.7 Rocket switch	23
4.2.8 Jumper wires	24
4.2.9 Battery Management System (3s 20A BMS)	25
4.2.10 3S 2A 18650 Lithium Battery Charger Board	26
Chapter 5: SOFTWARE IMPLEMENTATION	27
5.1 ESP32-CAM Using Arduino IDE	27
5.2 FTDI Adapter	29
5.3 AI-Powered Chatbot Web Application	30
Chapter 6: INTEGRATION AND TESTING	34
6.1 Integrating ESP32 – CAM and L298N Motor Driver	34
6.2 Integrating ESP32-CAM and MG996R Servo Motor	34
6.3 Connecting ESP32-CAM with Mobile	35
Chapter 7: RESULTS	38
7.1 Step-Wise Operational workflow of the Surveillance	38
7.2 Comparison Table	40
Chapter 8: CONCLUSION & FUTURE SCOPE	42
8.1 Conclusion	42
8.2 Future Scope	42
REFERENCES	43
APPENDIX-I	45
Sample Code	45

LIST OF FIGURES

Figure No.	Description	Page No.
Figure 1	Normal surveillance camera	1
Figure 2	Intelligent combat robot in a battlefield environment	5
Figure 3	Range-based navigation system for a mobile robot	6
Figure 4	Remote operated spy remote control system	7
Figure 5	Spherical robot named RollRoller	8
Figure 6	Spherical robot actuated by cylindrical rotor	9
Figure 7	MonoRollBot: 3-DOF spherical robot with under actuated single actuator	10
Figure 8	Block Diagram	12
Figure 9	Flowchart	13
Figure 10	ESP32 – CAM pinout configuration	16
Figure 11	L298N Motor Driver	17
Figure 12	Shaft Geared Motor	19
Figure 13	Buck Convertor (LM2596)	20
Figure 14	MG996R Servo Motor	21
Figure 15	18650 Li-ion Battery cell	22
Figure 16	Rocket switch	23
Figure 17	Jumper wires	24
Figure 18	Battery Management System	25
Figure 19	3S 2A 18650 Lithium Battery Charger Board	26
Figure 20	Arduino IDE	29
Figure 21	FTDI Adaptor	30
Figure 22	Web Application Interface	31
Figure 23	Web Application chatbot	31
Figure 24	Web Application chat response-1	32
Figure 25	Web Application chat response-2	32
Figure 26	Interfacing ESP32 – CAM and L298N Motor Driver	34
Figure 27	Interfacing ESP32-CAM and MG996R Servo Motor	35
Figure 28	Wi-Fi Connection Interface	36
Figure 29	Connecting to Wi-Fi Network	36
Figure 30	Entering IP Address to open Web Page	37

Figure 31	After Web Socket Connection Web Page	37
Figure 32	Wi-Fi Connection setup	38
Figure 33	Real time surveillance	39
Figure 34	Top view of Robot	39
Figure 35	Front view of Robot	40

LIST OF TABLES

Table No.	Description	Page No.
Table 1	Comparison table different research papers	41

Chapter 1

INTRODUCTION

1.1 Introduction

Surveillance operations are crucial in inaccessible war zones, during emergencies, and in hazardous areas where human intervention is unsafe. Robots can replace human efforts in such situations, offering remote operation capabilities. These spy robots are also essential for security in public buildings and during terrorist attacks. Challenges faced by human operators include bomb detection, monitoring unusual activities, border surveillance, patient monitoring, industrial oversight, and search-and-rescue missions during natural disasters.



Figure 1 Normal surveillance camera

Key functionalities required in a spy robot include object identification in unknown territories, live video streaming, and seamless mobility across diverse terrains. Traditional robots, such as legged or wheeled models, face limitations in speed, terrain adaptability, and visibility. A spherical design offers a superior alternative for inspection in areas inaccessible to humans. This design places a car based model within a spherical ball for enhanced performance. Research on propulsion mechanisms for spherical robots includes car-based models, independent hemispheres, and center-of-gravity relocation. The earliest spherical robot models employed wheels contacting the sphere's base, driven by a power and communication unit. While this design allowed for turning, it suffered from slow motion, instability upon collision, and internal slippage, resulting in imprecise movement.

Several advancements have been made in spherical spy robots. Remote controlled combat robots with laser guns and Advanced Technology for Memory and Logic (ATMEL) microcontrollers, as well as Programmable Interface Controllers (PIC)-controlled robots equipped with Charged-Coupled Device (CCD) cameras and Radio Frequency (RF) transmitters, have been developed.

Existing designs with perpendicular rotors, central bodies, and mass-shifting mechanisms offer improvements but are hindered by mechanical complexity, high motor performance demands, and limitations caused by central hubs. To address these challenges, the proposed spherical spy robot incorporates a pendulum-based drive system. This design provides live video streaming, obstacle detection, and stable locomotion using advanced control algorithms. The pendulum mechanism, powered by four 12V batteries, enables sphere steering by relocating the center of mass. A horizontal pipe structure supports the pendulum, ensuring stability and allowing for flexible sphere rotation and tilt.

1.2 Problem Statement

Modern robotic deployments face numerous challenges, particularly in adapting to uneven terrains and harsh environments. Traditional robots, including humanoid, wheeled, and legged designs, struggle with stability and mobility on rocky, slippery, or uneven surfaces, limiting their effectiveness in military surveillance, rescue missions, and other critical operations. Additionally, these robots require advanced sensors and control systems, leading to high production and operational costs. Communication barriers further hinder efficiency, as Bluetooth and GPS-based systems suffer from range limitations, reducing their capability for extensive surveillance and real-time data transmission. Addressing these challenges is crucial for enhancing robotic performance in demanding applications.

1.3 Objective

To design a spherical spy robot with a pendulum-based drive system for simplified structure enhanced mobility through center of mass adjustment, and real time video streaming for live surveillance.

1.4 Methodology

The developed system operates in real-time, designed as a spherical spy robot with a pendulum-based drive mechanism for enhanced mobility and stability. The robot

navigates diverse terrains by adjusting its center of mass, enabling smooth rotation and tilt. It is equipped with a real-time video streaming module to capture live surveillance footage. Obstacle detection sensors ensure safe navigation, while the control system processes movement data for precise manoeuvrability. The video feed is transmitted to a remote device via wireless communication, allowing operators to monitor activities. A suitable control algorithm optimizes motion and balance, ensuring efficient surveillance performance.

1.5 Motivation

The motivation for the development of a spherical spy robot with a pendulum based drive system is driven by the need to enhance robotic capabilities for critical operations in challenging environments. Traditional robots face significant limitations when navigating such as rocky or slippery surfaces, reducing their effectiveness in military surveillance, rescue missions, and other high-stakes applications. Additionally, existing systems suffer from high costs due to complex sensors and control systems, along with communication barriers that restrict range and real-time data transmission.

1.6 Thesis Organization

Chapter 1: This chapter consists of introduction of our project, problem statement, objective of the project, methodologies used and contribution of the project.

Chapter 2: Literature survey done by previous work.

Chapter 3: This chapter consists of problem specifications. In this chapter we have given detailed explanation of block diagram.

Chapter 4: In this chapter, a detailed explanation is provided for the hardware components used in the project.

Chapter 5: This includes description about the software used in our project.

Chapter 6: This consists of integration and testing which describes in detail about hardware components integration with one another.

Chapter 7: This chapter consists of results of our project.

Chapter 8: Summarizing and giving the observations of project and giving the reason of its implementation.

Chapter 2

LITERATURE SURVEY

2.1 Introduction

Spherical robots have emerged as a promising solution for surveillance operations due to their unique mobility, structural robustness, and ability to navigate diverse terrains with minimal external exposure. Unlike traditional wheeled or legged robots, spherical robots encapsulate all components within a rolling shell, offering superior protection and omnidirectional movement. Recent advancements in control systems, sensor integration, and communication technologies have significantly enhanced their operational capabilities in real-time monitoring, reconnaissance, and security tasks. This literature survey explores the development, design methodologies, and application-specific innovations in spherical robots tailored for surveillance purposes..

2.2 Literature Reviews

[1] Q. Jia, H. Sun, and D. Liu, "Analysis of actuation for a spherical robot," Proc. 2008 IEEE Conf. Robot. Autom. Mechatronics, pp. 1–6, 2008. [1]

The paper titled "Analysis of Actuation for a Spherical Robot" by Q. Jia, H. Sun, and D. Liu, presents a comprehensive study on the actuation mechanisms of spherical robots. The key findings of this research include the development of a dynamic model for the spherical robot, incorporating the effects of a counter-weight pendulum attached to a gimbal, a steering motor, and a drive motor. This model facilitates the analysis of the robot's motion equations and non-holonomic constraints, providing insights into the robot's manoeuvrability and control. The study also includes simulation studies to validate the proposed model, demonstrating its effectiveness in predicting the robot's behaviour under various conditions.

The main challenge in this project was the reliance on a counter-weight pendulum and gimbal system may introduce mechanical complexity, potentially affecting the robot's reliability and maintenance requirements. Additionally, the dynamic model assumes ideal conditions, which may not fully account for real-world factors such as friction, sensor inaccuracies, and environmental disturbances. The simulation studies, while valuable, may not capture all the complexities of physical implementation, necessitating further empirical testing to validate the model's predictions.

[2] S. Bhargavi and S. Manjunath, "Design of an intelligent combat robot for war fields," 2011 International Journal of Computer Applications, vol. 2, no. 8, pp. 13-17, 2011. [2]

The "Design of an Intelligent Combat Robot for War Fields" project presents several key findings that highlight its potential for enhancing military and security operations. Its radio-controlled design facilitates remote operation, while its stealth capabilities enable infiltration into enemy zones for surveillance and reconnaissance. Additionally, the robot is self-powered, ensuring sustained operation during extended missions.

However, the project has certain limitations. The robot's payload capacity is restricted, limiting the tools and equipment it can carry, which could hinder its use in more complex operations. Terrain navigation could be a challenge, as the robot may struggle in rough or difficult environments. Its reliance on wireless communication makes it vulnerable to jamming or interference, and the battery life may constrain its operational duration.

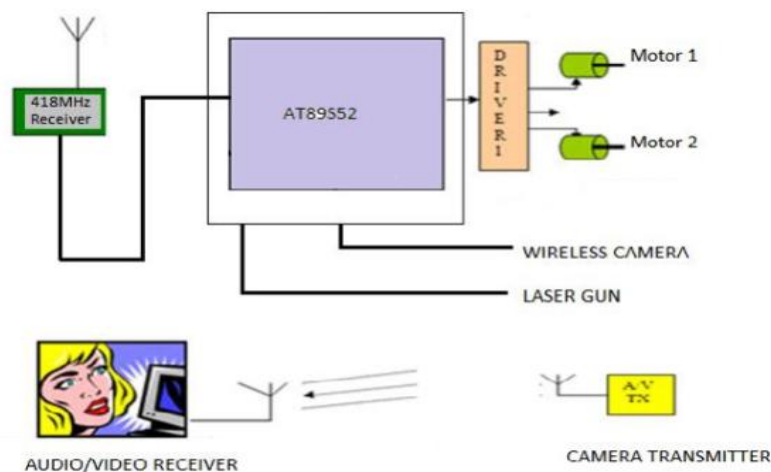


Figure 2 : intelligent combat robot in a battlefield environment

[3] N. MacMillan, R. Allen, D. Marinakis, and S. Whitesides, "Range-based navigation system for a mobile robot," Proc. 2011 Canadian Conf. Comput. Robot. Vis., pp. 16–23, May 2011. [3]

The project titled "Range-based Navigation System for a Mobile Robot" by N. MacMillan, R. Allen, D. Marinakis, and S. Whitesides, focuses on developing an algorithm for path planning in environments equipped with fixed range-only beacons. The key findings of this research include the introduction of a method to define and calculate entropy values for regions of interest within the environment. By assessing

these entropy values, the algorithm identifies "safe," low-entropy paths between regions, facilitating efficient navigation for mobile robots.

It is important to acknowledge the limitations of this project, particularly the reliance on range-only beacons may limit the robot's ability to detect obstacles or navigate in environments where beacon signals are weak or obstructed. Additionally, the system's performance is dependent on the accuracy and placement of the beacons, which may not be feasible in all environments. The use of acoustic beacons, while cost-effective, may be susceptible to interference from environmental noise, potentially affecting the reliability of the ranging measurements. Furthermore, the algorithm's effectiveness in dynamic environments, where obstacles and conditions change over time, remains uncertain and would require further investigation.

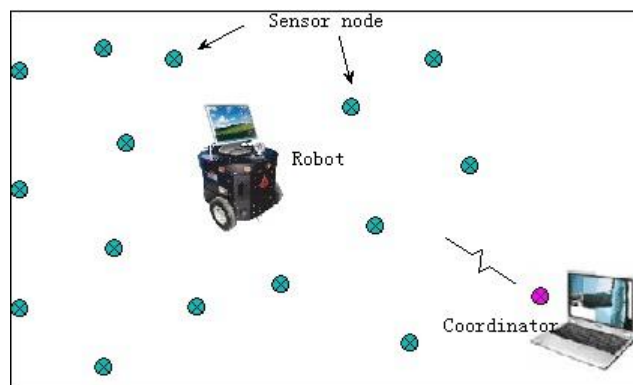


Figure 3 Range-based navigation system for a mobile robot

[4] W. M. M. Khaing and K. Thiha, "Design and implementation of remote operated spy robot control system," *Int. J. Adv. Res. Electr. Electron. Instrum. Eng.*, vol. 3, no. 7, pp. 3545-3550, Jul. 2014. [4]

The project "Design and Implementation of Remote Operated Spy Robot Control System" introduces a versatile and compact robot designed for remote surveillance and inspection in hazardous environments. The robot uses two microcontrollers: PIC 16F628A for remote control operations and PIC 16F877 for managing the robot's functions. The robot is powered by three brush DC motors controlled by L298N motor drivers, providing smooth and responsive movement. Additionally, the system features an LCD display on the remote control unit to display user commands, ensuring intuitive operation. The use of Radio Frequency (RF) modules for wireless communication enables reliable control of the robot over short distances, making it suitable for use in

surveillance, reconnaissance, and inspection in areas like industrial sites, disaster zones, and military operations.

Despite its promising design, the project has certain limitations. The robot's reliance on RF communication means it may experience interference or range limitations in environments with high electromagnetic noise or physical obstructions. The control system is designed for short-range use, which could hinder the robot's effectiveness in larger or more complex environments. Another limitation is the relatively simple nature of the camera system, which, while effective for basic surveillance, may not offer high-definition quality or advanced features like zoom or pan/tilt capabilities. The use of basic brush DC motors also limits the robot's manoeuvrability, especially in rugged or uneven terrains, where more advanced motor systems could be beneficial.

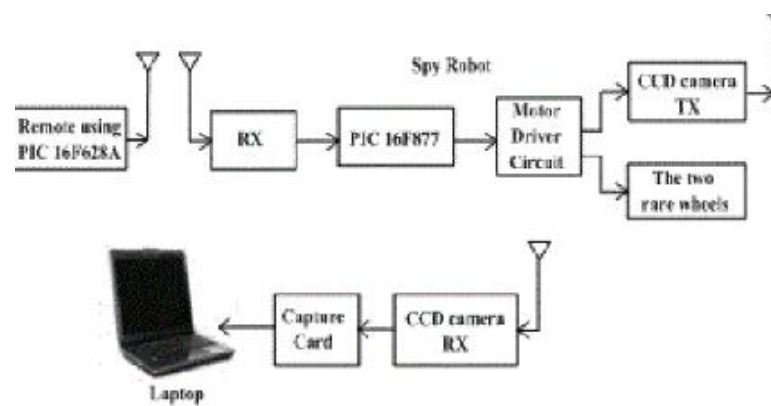


Figure 4 Remote operated spy remote control system

[5] G. Vashisht and R. Dhod, "Defence surveillance robot based on Radio Frequency (RF) and Dual-Tone Multi-Frequency (DTMF) technology," *Int. J. Adv. Res. Electr. Electron. Instrum. Eng.*, vol. 4, no. 6, pp. 5469-5473, Jun. 2015.[5]

The project "Surveillance Based on DTMF Technology" focuses on the development of a defense surveillance robot that integrates RF and DTMF technologies for enhanced monitoring and control in hazardous environments. The key findings of this project include the use of advanced sensors such as metal detectors, fire sensors, and infrared sensors to enable the robot to perform a range of tasks, from surveillance to fire detection and explosive disposal. The robot's operational flexibility is one of its standout features, with three modes of operation: RF mode, DTMF mode, and automatic mode. RF mode allows for control over short distances, DTMF mode enables long-range control via GSM mobile phones, and the automatic mode utilizes sensors to function autonomously.

A key constraint faced during this project was the robot's performance heavily relies on wireless communication, which could be vulnerable to interference or jamming, especially in combat or hostile environments. The RF range of the robot is limited to approximately 200 meters, which might restrict its operational scope in larger areas or more complex terrains. Additionally, while the DTMF mode allows for long-range control, it is still dependent on GSM network availability, which may not always be reliable in remote or conflict zones.

[6] P. S. Li, R. Z. Zhang, and X. L. Chen, "Dynamical behavior investigation and analysis of novel mechanism for simulated spherical robot named 'RollRoller,'" *J. Dyn. Robot.*, vol. 8, no. 4, pp. 210-215, 2016. [6]

The RollRoller robot is designed with a fluid-actuated multi-driven closed system, which offers the robot the ability to move smoothly and efficiently across a variety of surfaces. The design allows the robot to exhibit excellent stability and control, even in situations requiring complex movements or precise navigation. The research presents an in-depth analysis of the robot's dynamics and how it interacts with different types of surfaces.

The fluid-actuated system could be prone to failures or inefficiencies if the fluid supply is disrupted or if the system experiences leaks. Additionally, the system's complexity might increase the cost and maintenance efforts.

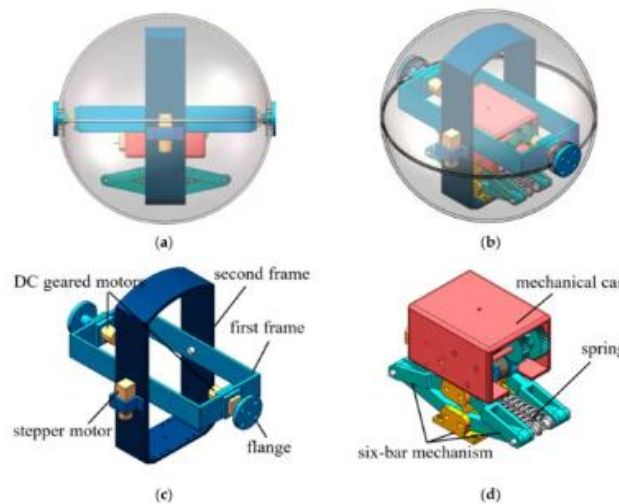


Figure 5 Spherical robot named RollRoller

[7] L. C. Yang, K. H. Tan, and J. P. Lee, "Design and modeling of a spherical robot actuated by a cylindrical rotor," *IEEE Robot. Autom. Mag.*, vol. 27, no. 2, pp. 45-50, 2020. [7]

This paper explores a spherical robot design actuated by a cylindrical rotor, which allows for precise movement in three dimensions. The unique aspect of the robot is its method of propulsion, where the cylindrical rotor's rotation drives the robot's movement, providing a high degree of manoeuvrability and stability. The research presents a detailed dynamic model to simulate the robot's behaviour and performance under various conditions.

Although the cylindrical rotor design provides effective control, it may still face challenges with traction and stability on rough or sloped surfaces. Additionally, the complexity of the rotor system may increase maintenance demands. Introducing a pendulum-based system could help enhance the robot's balance, providing a counteracting force that compensates for the potential instability caused by the rotor's rotational forces.

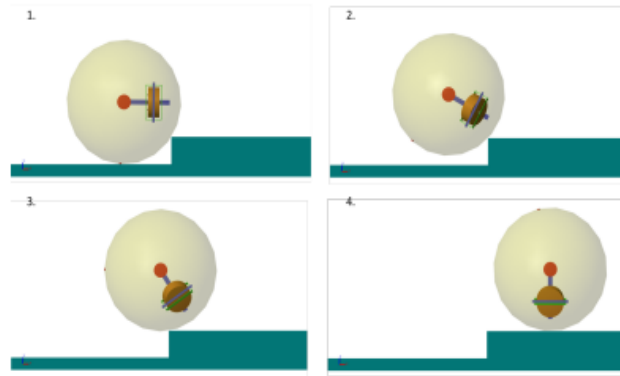


Figure 6 Spherical robot actuated by cylindrical rotor

[8] M. A. Khan and A. B. Taufik, "MonoRollBot: 3-DOF spherical robot with under actuated single actuator," *Proc. 2021 IEEE Int. Conf. Robot. Autom.*, pp.1234-1239, 2021.[8]

The MonoRollBot is a spherical robot that operates with a single actuator, enabling it to move in three degrees of freedom (3-DOF). The design utilizes an under actuated mechanism where the robot's movement is controlled by a single point of actuation, making it highly efficient and lightweight. The key achievement of the MonoRollBot is its ability to move across complex terrains while maintaining stability, thanks to its innovative internal actuation mechanism, which allows the robot to roll with minimal external input.

One limitation of the MonoRollBot is that its stability could be compromised when encountering uneven terrain or external disturbances due to the simplicity of its actuation system. The lack of additional actuators might result in difficulties with precise manoeuvring in dynamic environments. A pendulum-based system could be integrated to enhance the robot's balance and improve its performance on unstable surfaces by using the pendulum as a counterbalance to the robot's motion

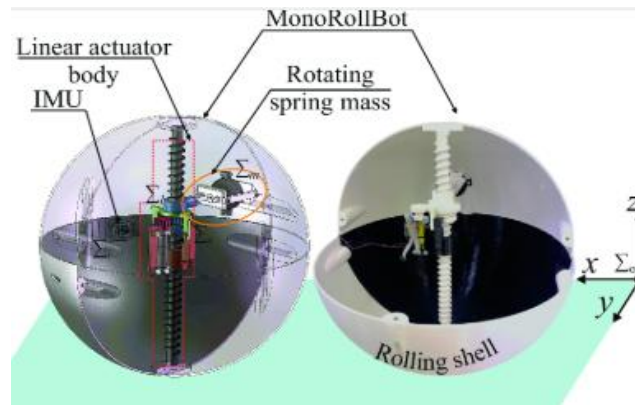


Figure 7 MonoRollBot: 3-DOF spherical robot with underactuated single actuator

[9] S. T. Williams, M. K. Patel, "Design and control of a multi-DOF spherical robot for autonomous exploration," *IEEE Trans. Robot. Autom. Syst.*, vol. 40, no. 5, pp. 455-462, 2022. [9]

This paper presents a multi-degree-of-freedom (DOF) spherical robot designed for autonomous exploration tasks, specifically in unknown and rugged environments. The system is equipped with an efficient control algorithm to enable precise movement and obstacle navigation. The robot's motion control strategy focuses on using omni-wheels and a central control system that integrates vision-based feedback for real-time obstacle detection.

One limitation of the design is the reliance on vision-based sensors, which can struggle in low-light or occluded environments, reducing the robot's overall autonomy. Additionally, the multi-DOF configuration increases the complexity of the control algorithms, requiring substantial computational resources. To mitigate these issues, adding supplementary sensors like LIDAR or ultrasonic sensors could enhance the robot's robustness in challenging environments.

[10] L. K. Lee and T. N. Mishra, "Spherical robot dynamics and motion planning," *IEEE Int. Conf. Robot. Autom.*, pp. 901-907, 2023. [10]

The paper explores the dynamics of spherical robots, especially those utilizing pendulum actuators for stabilization and control. The research investigates the motion planning algorithms that optimize the movement of spherical robots while minimizing the need for external control. By incorporating pendulum-based actuators, the robot's center of gravity can be more effectively managed, leading to improved stability and precise navigation in complex environments. The study provides a detailed analysis of the robot's movement trajectories and their impact on the robot's performance. One key limitation is that while the pendulum-based actuator contributes to stability, it introduces a dependency on the precise tuning of the pendulum's mass and length. Any slight misalignment can lead to instability or difficulty in maneuvering the robot. A solution to overcome this challenge could be the incorporation of a feedback control system that adjusts the pendulum's properties in real time based on sensory inputs, enhancing the robot's adaptability to dynamic environments.

Chapter 3

PROPOSED SYSTEM

3.1 Introduction

The Pendulum Driven Spherical Spy Robot is designed to enhance surveillance operations with a spherical structure driven by an internal pendulum mechanism, allowing for 360-degree motion and stealthy movement across various terrains. It addresses the limitations of traditional surveillance robots by providing noiseless, energy-efficient mobility and real-time monitoring through integrated camera.

3.2 Block Diagram

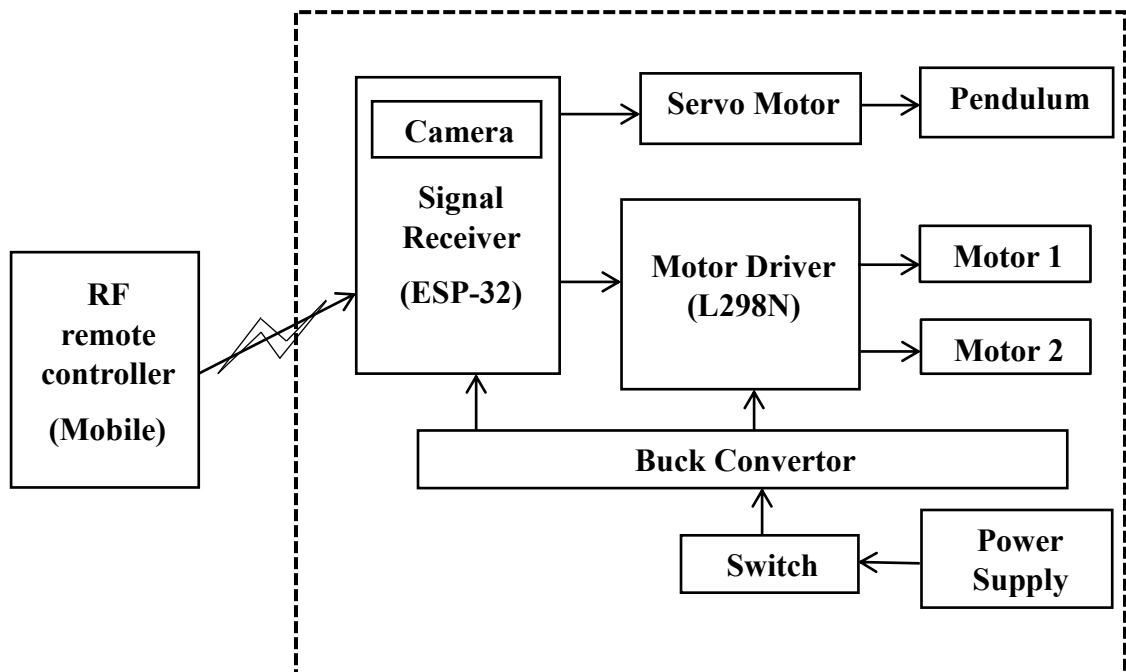


Figure 8 Block Diagram

3.3 Methodology

The program in ESP 32 interacts with hardware and software environment is through serial data communication. The details of operation are shown in Figure 8. With the connection of Bluetooth through a specific IP address, the mobile application is activated and the rotation angle for the pendulum is set to 90 degree using servo motor. With fixing angle of pendulum, forward button in mobile application is used to move the spherical spy robot in the forward direction. With the forward and reverse movement, DC motor

rotates in clockwise and anticlockwise respectively. To terminate the process, stop button in the mobile application used to stop the rotation of DC motor.

3.4 Flowchart

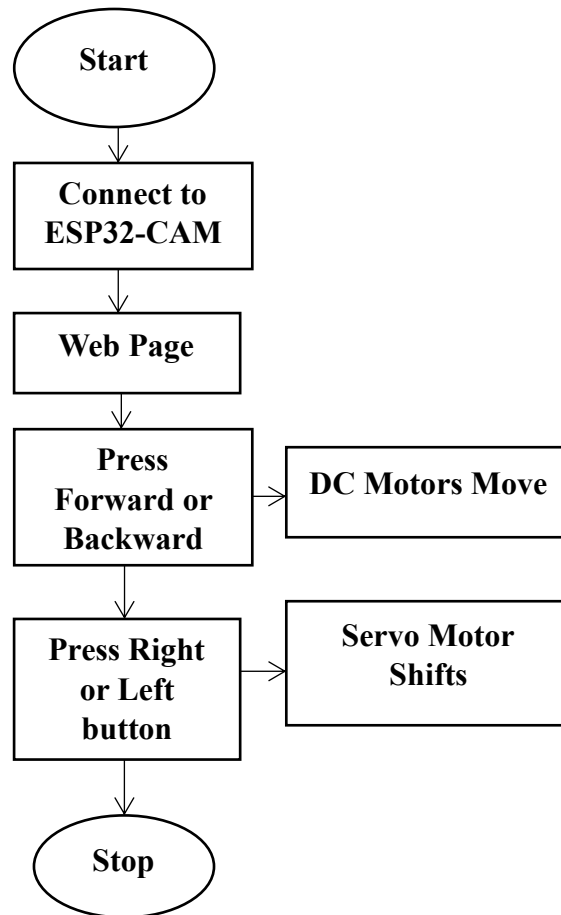


Figure 9 Flowchart of the Robot Program

3.5 Applications

1. Environmental Monitoring

- **Wildlife Surveillance:** Spherical robots can quietly navigate diverse terrains (forest floors, wetlands) to monitor animals without disturbing them.
- **Pollution Detection:** Equipped with sensors for air, water, or soil quality, they can monitor pollutants in hard-to-reach or hazardous areas.

2. Agriculture

- **Crop Surveillance:** Roll through fields capturing visual and thermal data to detect pest infestations, drought stress, or disease.

- **Soil Monitoring:** Sensors can collect data on moisture, pH, and nutrient levels.

3. Healthcare

- **Elderly and Patient Monitoring:** In smart healthcare environments, spherical robots can patrol rooms, detect falls, or alert caregivers of abnormal activity.
- **Sanitation Surveillance:** Monitor hygiene practices and surface cleanliness in hospitals.

4. Home Automation

- **Security Surveillance:** Patrol homes autonomously, equipped with cameras and motion detectors. Can send alert if unusual activity is detected.
- **Pet Monitoring:** Monitor pets while owners are away and interact with them using speakers or treat dispensers.

5. Defence

- **Border Surveillance:** Deploy in rough terrain for continuous patrolling and intruder detection.
- **Urban Combat/Reconnaissance:** Enter buildings or narrow passages to provide live feeds before troops move in.

3.6 Advantages

1. Human Friendly: Safe for use around people due to its smooth, enclosed design.
2. Eco Friendly: Low power use and minimal environmental impact.
3. Adaptability: Works well across various terrains and tight spaces.
4. Manoeuvrability: Moves easily in all directions with simple external design.

3.7 Disadvantages

1. Capacity: Limited space for heavy sensors or equipment.
2. Complexity: Involves complicated design and control system.

Chapter 4

HARDWARE IMPLEMENTATION

4.1 Introduction

The hardware implementation of the project involves a pendulum-driven spherical spy robot for surveillance. The goal is to create a small and discreet robot using a mix of electronic and mechanical parts. Its round shape allows it to move in any direction, making it easy to navigate tight or complex spaces.

The robot moves by using a weight inside called a pendulum. When the pendulum shifts, it makes the round robot roll that way. We can control this movement using Wi-Fi, so no wires are needed.

Key parts of the robot include an ESP32-CAM for live video, servo motors for moving the pendulum, Shaft metal Geared motors, a motor driver (L298N), a buck converter for managing power, a battery, and a Battery Management System (BMS). All these parts fit into a compact round body for better balance.

4.2 Hardware selection

- ESP32 - CAM
- L298N Motor Driver
- Shaft Metal Geared Motors
- Buck Convertor (LM 2596)
- MG996R Servo Motor
- 12V Battery
- Rocker Switch (SPST)
- Jumper wires
- Battery Management System (3S 20A BMS)
- 3S 2A 18650 Lithium Battery Charger Board USB C

The hardware components are explained below

4.2.1 ESP32 – CAM

The ESP32-CAM is a small-sized, low-cost development board that integrates an ESP32-S chip and a camera module. It is a versatile board that combines Wi-Fi and Bluetooth connectivity with image processing capabilities, making it ideal for various Internet of Things (IoT) applications. Common use cases include surveillance cameras, facial recognition systems, home automation, and remote monitoring.

Its standout feature is the ability to capture and process images in real time, enabling applications such as surveillance systems, facial recognition, home automation, and remote monitoring. The board is especially useful in security setups where discreet and low-power cameras are needed, or in smart home devices like video doorbells and pet monitors.

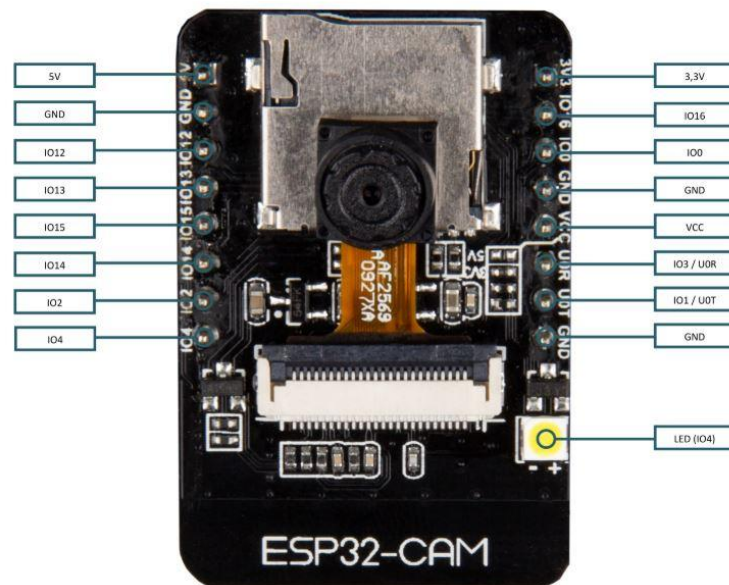


Figure 10 ESP32 – CAM pin out configuration

Specifications

- Chipset: ESP32-S dual processor
- RAM: 520 KB SRAM
- Flash Memory: 4 MB
- Wi-Fi: 802.11b/g/n
- Bluetooth: v4.2 BR/EDR and BLE
- Camera Interface: Supports OV2640 and OV7670 cameras
- I/O: GPIO, UART, SPI, I2C
- Operating Voltage: 5V
- Operating Temperature: -20°C to 85°C

Pinout Configuration

- **Power Supply:** Ensure that the ESP32-CAM is powered with a stable 5V supply. An unstable or incorrect voltage can damage the board.

- **Camera Connection:** Attach the camera module to the ESP32-CAM board carefully, ensuring proper alignment and connection.
- **Programming:** The ESP32-CAM does not come with a USB port, so you will need an FTDI programmer to upload code.
- **GPIO Access:** Use the I/O pins for connecting peripherals, but be aware of the pins used by the camera and other on-board features to avoid conflicts.
- **GND:** Ground pins.
- **5V:** 5V power supply.
- **U0TXD:** UART0 Transmit.
- **U0RXD:** UART0 Receive.

4.2.2 L298N Motor Driver

L298N is a high current-high voltage dual full-bridge driver manufactured to operate inductive loads like stepping motors, relays. It also receives standard TTL (Transistor-Transistor Logic) logic signals. L298N is an integrated monolithic circuit in PowerSO20 and 15-lead watt packages. It has two enable inputs that independently disable or enable a device attached at its output irrespective of input signals. Additionally, it regulates the rotating direction in DC motors. Its function is directly controlling a DC (two brushed) or Stepper motor (one 4-wire two-phase). At it's centre,

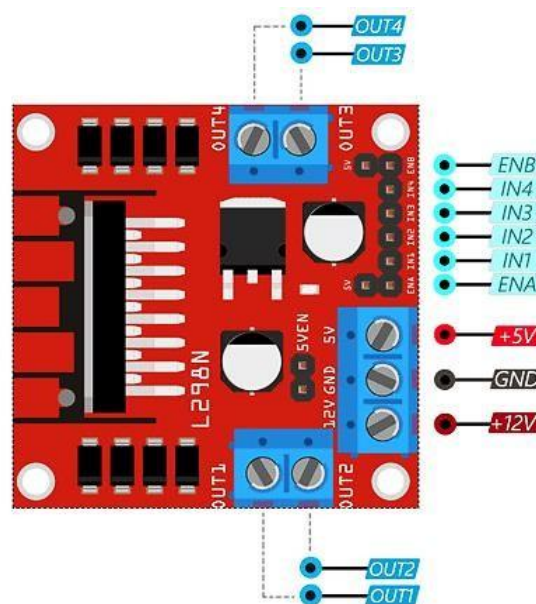


Figure 11 L298N Motor Driver

There's a dual channel H-bridge driver chip capable of withstanding approximately 2A current per channel. An Atmel ATmega8L controls the chip and can manage I2C interfaces with platforms like Arduino. It can also simultaneously drive the stepper and DC motor in different directions and speeds. L298N motor driver is used for controlling the movement of DC motors in various applications, such as wheeled robots, robotic arms, and drones. Its bidirectional control capability allows robots to move forward, backward, turn left, turn right, and perform other manoeuvres. It's often paired with microcontrollers like Arduino, Raspberry Pi, or other control systems to receive commands and drive the motors accordingly.

Specifications

- Driver Voltage: 5-35V
- Logic Voltage: 5V TTL Compatible
- Driver Current: 2A
- Operating Temperature: -25 °C TO +130°C

Features

- Dual H-Bridge motor driver
- Direction and speed control
- Heat sink for better performance
- Thermal shutdown
- Overcurrent Protection

Applications

- Remote Controlled Vehicles
- Electronic Door Locks
- CNC Machines

4.2.3 Shaft Geared Motor

The Center Shaft Metal Geared Motor is a robust and versatile motor designed to deliver high torque and precise control. Equipped with a metal gearbox and a central shaft, this motor is ideal for applications requiring reliable and efficient mechanical movement. Common use cases include robotics, automation systems, conveyor belts, and other machinery where precise control and high torque are essential.

This motor is commonly used in robotics, where its high torque enables it to drive wheels, arms, or joints with accuracy and strength, making it essential for mobile robots,

robotic arms, and automated guided vehicles (AGVs). In automation systems, the center shaft motor plays a key role in driving actuators, levers, or mechanisms that require synchronized and repeatable movement. It is also widely employed in conveyor belt systems, where consistent torque is needed to move materials or products smoothly and reliably. Beyond industrial applications, these motors are frequently found in DIY projects, educational kits, and mechatronics prototypes, offering a dependable solution for students and engineers alike.



Figure 12 Shaft Geared Motor

Specifications

- Operating Voltage: 6V - 12V, Rated Current: 0.5A - 1.2A
- Shaft Diameter: 6mm
- Shaft Length: 20mm
- Weight: 150g

Pinout Configuration

- VCC: Power supply (6V - 12V), GND: Ground
- IN1: Motor control input 1
- IN2: Motor control input 2

Applications

- Industrial machinery
- Robotics
- Security systems

4.2.4 Buck Convertor (LM2596)

The LM2596, manufactured by STMicroelectronics (Part ID: UNO), is a step-down (buck) voltage regulator designed to efficiently convert a higher input voltage to a lower

output voltage. It is widely used in power supply applications due to its high efficiency, ease of use, and robust protection features. The LM2596 can handle up to 3A of output current and includes built-in thermal shutdown and current limiting to ensure safe operation.



Figure 13 Buck Convertor (LM2596)

Specifications

- Input Voltage Range: 4.5V to 40V , Output Voltage Range: 1.23V to 37V
- Maximum Output Current: 3A, Efficiency: Up to 92%
- Switching Frequency: 150 kHz , Output Voltage Tolerance: $\pm 4\%$
- Operating Temperature Range: -40°C to $+125^{\circ}\text{C}$
- Protection Features: Thermal shutdown, current limiting.

Pin out Configuration

- VIN: Input voltage pin. Connect to the unregulated DC input voltage.
- Output: Regulated output voltage pin. Connect to the load.
- Ground (GND): Ground pin. Connect to the system ground.
- Characteristics: CYC-Serie: Cycle
- Feedback: Feedback pin. Used to set the output voltage via an external resistor divider.
- ON/OFF: Enable pin. Pull high to enable the regulator, or low to disable it.

Applications

- Voltage regulation for microcontrollers and sensors
- DC-DC power supply modules for embedded systems
- Battery-powered devices

4.2.5 MG996R Servo Motor

MG996R servo motor is a high-torque and precise motor ideal for robotics and automation projects. With its durable construction, strong torque of up to 11 kg/cm, and quick response time, it provides reliable and accurate motion control. The MG996R servo motor also supports continuous rotation modification, providing flexibility for specific applications. Its compatibility with various control systems and microcontrollers makes integration into your project hassle-free. The motor comes with a standard 3-pin connector for easy wiring and is designed for straightforward plug and-play functionality.

Specifications

- Operating Voltage is +5V typically , Current: 2.5A (6V)
- Stall Torque: 9.4 kg/cm (at 4.8V) , Maximum Stall Torque: 11 kg/cm (6V)
- Gear Type: Metal , Rotation : 0-180 degrees
- Weight of motor : 55gm
- Package includes gear horns and screws



Figure 14 MG996R Servo Motor

Wire Configuration

- Brown: Ground wire connected to the ground of system
- Red: Powers the motor typically +5V is used
- Orange: PWM signal is given in through this wire to drive the motor

Features

- Metal gears
- Double ball bearing design
- High Torque
- Wide voltage range
- Standard PWM Control

Applications

- Used as actuators in many robots like Biped Robot, Hexapod, robotic arm
- Commonly used for steering system in RC toys
- Home Automation Systems
- Less weight hence used in multi DOF robots like humanoid robots

4.2.6 12V Battery (18650 Li-ion Battery)

The 18650 Li-ion Battery is a standard size rechargeable cylindrical cell, renowned for its balance between energy capacity and power delivery. This battery type is widely used in various electronic devices such as laptops, flashlights, portable power banks, and even in electric vehicles due to its high energy density and longevity. Its name, "18650," refers to its dimensions: approximately 18mm in diameter and 65mm in length.

Three 18650 Li-ion batteries, each rated at 3.7V, are connected in series to achieve a combined output voltage of approximately 11.1V to 12.6V. This series configuration sums the individual cell voltages, providing a total nominal voltage of 11.1V ($3.7V \times 3$), with a fully charged voltage reaching up to 12.6V ($4.2V \times 3$), making it suitable for powering 12V systems.



Figure 15 18650 Li-ion Battery cell

Specifications

- Nominal Voltage : 3.6V - 3.7V
- Fully Charged Voltage: 4.2V
- Discharge Cut-off Voltage: 2.5V - 3.0V
- Typical Capacity: 1800mAh - 3500mAh (varies by manufacturer)
- Max Charging Current: 0.5C - 1C (C = Battery Capacity in Ah)
- Max Discharge Current: Varies by model and manufacturer.
- Chemistry: Lithium-ion
- Cycle Life: Typically 300-500 cycles.

Pin out Configuration

- Positive (+): The raised terminal on the top of the battery.
- Negative (-): The flat terminal on the bottom of the battery.

4.2.7 Rocket switch

A rocket switch, also known as a launch switch or ignition switch, is a type of electrical switch used to initiate the launch sequence of a rocket or other aerospace vehicle. It is a critical component of the launch control system and is typically designed to withstand the harsh conditions of space travel. Rocket switches are often designed with safety features to prevent accidental activation and ensure reliable operation during the launch sequence. They may include locking mechanisms or require multiple steps to activate, such as a sequence of switches or buttons that must be pressed in a specific order. In addition to aerospace applications, similar switches may also be used in other high-reliability and safety-critical systems, such as military equipment or industrial machinery where precise control and safety are paramount.



Figure 16 Rocket switch

Features

- Reliability: Highly dependable for critical aerospace operations
- Safety: Equipped with measures to prevent accidental activation
- Environmental Resistance: Resistant to moisture, dust, and other environmental factors

Applications

- Household appliances , Electrical equipment

- Industrial and domestic appliances

4.2.8 Jumper wires

Jumper wires are a fundamental aspect of electronics prototyping and circuit building. When connecting jumper wires, start by selecting the appropriate wires for your project. Identify the connection points on your breadboard or circuit board that you want to link. Then, strip a small amount of insulation from the ends of the wires if needed. Insert one end of the wire into the connection point, ensuring it's securely seated. Next, connect the other end of the wire to the desired destination, such as another connection point, pin, or terminal. Repeat this process for each connection needed in your circuit. Finally, test the circuit to ensure proper connectivity and functionality.

Specifications

- Jumper wire size: 26 AWG 31 , Current Rating: up to 1 A , Insulation Type: PVC



Figure 17 Jumper wires

Features

- Versatility: Jumper wires are versatile tools for making electrical connections.
 - Flexibility: These wires are flexible, allowing easy routing within circuits
 - Availability: They come in various lengths, colors, and connector types
 - Customization: Can be cut to custom lengths if needed
 - Reusable: Easily disconnected and reconnected for prototyping
 - Compatibility: Work with standard breadboards and electronic components □
- Rapid Prototyping: Ideal for quick experimentation and circuit creation.

Applications

- Prototyping
- Automotive Electronics

- Telecommunication

4.2.9 Battery Management System (3s 20A BMS)

The 3s 20A BMS (Battery Management System) is an essential circuit component designed for managing a 3-cell series lithium-ion battery pack. It ensures the longevity and safety of the battery pack by providing cell balancing, overcharge protection, over-discharge protection, and short-circuit protection. This BMS is capable of handling a maximum continuous discharge current of 20A, making it suitable for various applications such as DIY power banks, portable electronics, and small electric vehicles.

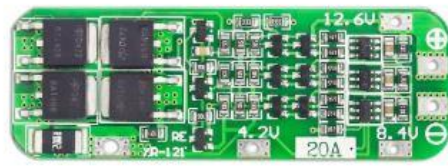


Figure 18 Battery Management System

Specifications

- Cell Count: 3 Series (3s) , Continuous Discharge Current: 20A
- Charge Voltage: 12.6V (4.2V per cell)
- Overcharge Protection: >4.25V per cell
- Over-discharge Protection: <2.5V per cell
- Short-Circuit Protection: Active

Pin out Configuration

- Battery Positive (B+): Connect to battery positive
- Battery Negative (B-): Connect to battery negative
- Load/Charger Positive (P+): Connect to load/charger positive
- Load/Charger Negative (P-): Connect to load/charger negative

Applications

- DIY power banks , Portable electronic devices
- Electric scooters and bicycles , Small-scale energy storage systems

4.2.10 3S 2A 18650 Lithium Battery Charger Board USB C

The 4S 2A 18650 Lithium Battery Charger Board USB C (Manufacturer: Arduino, Part ID: 4S 2A C-Type) is a compact and efficient charging module designed for charging four 18650 lithium-ion batteries connected in series. It supports a maximum charging

current of 2A and features a USB-C input for modern and convenient power delivery. This board is ideal for applications requiring a reliable and safe charging solution for high-capacity battery packs.

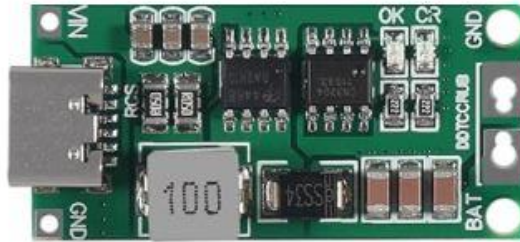


Figure 19 Battery Management System

Specifications

- Input Voltage: 5V (via USB-C port) , Charging Current: Up to 2A
- Battery Configuration: 3S (four 18650 cells in series)
- Battery Voltage Range: 14.8V nominal (16.8V fully charged)
- Protection Features: Overcharge, over-discharge, and short-circuit

Pin out Configuration

- USB-C Port: Input power supply (5V) for charging the battery pack.
- B+: Positive terminal for the first battery in the series,
- B1, B2, B3: Intermediate connections for balancing.
- P+: Positive output terminal for the connected load.
- P-: Negative output terminal for the connected load.

Applications

- Power banks and portable energy storage systems.
- DIY electronics projects requiring a 4S lithium battery pack.
- Robotics and remote-controlled devices.
- Solar energy storage systems.
- Backup power supplies for embedded systems.

Chapter 5

SOFTWARE IMPLEMENTATION

5.1 ESP32-CAM Using Arduino IDE

An integrated development environment (IDE) is a software application that provides comprehensive facilities to computer programmers for software development. Arduino IDE is open-source software that is mainly used for writing and compiling the code into the Arduino Module. An IDE normally consists of at least a source code editor, build automation tools, and a debugger. The main code, also known as a sketch, created on the IDE platform will ultimately generate a Hex File which is then transferred and uploaded in the controller on the board. The IDE environment mainly contains two basic parts: Editor and Compiler where former is used for writing the required code and later is used for compiling and uploading the code into the given Arduino Module. This environment supports both C and C++ languages.

The ESP32-CAM is a low-cost development board with an onboard camera, Wi-Fi, and Bluetooth. This document guides you through setting up the Arduino IDE to program the ESP32-CAM module, wiring it correctly, uploading code, and viewing the camera stream.

Required Components

- ESP32-CAM module (AI-Thinker)
- USB to TTL Converter (e.g., FTDI)
- Computer with Arduino IDE

Step 1: Install Arduino IDE

- Download from <https://www.arduino.cc/en/software>
- Install for your OS (Windows, macOS, or Linux)

Step 2: Install ESP32 Board Support

- Open Arduino IDE
- Go to File > Preferences
- Paste this URL into 'Additional Board Manager URLs': https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
- Go to Tools > Board > Board Manager, search for 'ESP32' and install it.

Step 3: Wiring ESP32-CAM to FTDI Adapter

Connect the following:

- GND to GND
- 5V to VCC
- U0R to TX
- U0T to RX
- IO0 to GND (for upload mode)

Note: Disconnect IO0 from GND after uploading.

Step 4: Select Board in Arduino IDE

- Tools > Board > ESP32 Arduino > AI Thinker ESP32-CAM
- Upload Speed: 115200
- Partition Scheme: Huge App
- Select correct COM port

Step 5: Load and Modify Example Code

- File > Examples > ESP32 > Camera > CameraWebServer
- Set your WiFi credentials
- `const char* ssid = "Your_SSID";`
- `const char* password = "Your_PASSWORD";`
- Enable the correct camera model.
- `#define CAMERA_MODEL_AI_THINKER`

Step 6: Upload Code

- Connect IO0 to GND.
- Click Upload.
- When 'Connecting...' appears, press the RESET button.
- After upload, disconnect IO0 from GND and press RESET again.

Step 7: View Camera Feed

- Open Serial Monitor (115200 baud)
- Look for IP address
- Open IP in browser to access live camera stream

Tips & Troubleshooting

- Double-check RX and TX wiring.

- Use stable 5V power supply.
- If upload fails, press RESET during 'Connecting...'
- Ensure router assigns IP addresses (DHCP).



Figure 20 Arduino IDE

5.2 FTDI Adapter

An FTDI adapter is a compact electronic device used to convert USB signals to serial communication signals (UART), enabling modern computers to interface with devices that use serial communication protocols. It is built around a chip developed by FTDI (Future Technology Devices International), most commonly the FT232RL. These adapters are essential for programming and debugging microcontrollers, development boards like the Arduino Pro Mini, or Wi-Fi modules such as the ESP8266 and ESP32, which often lack built-in USB connectivity.

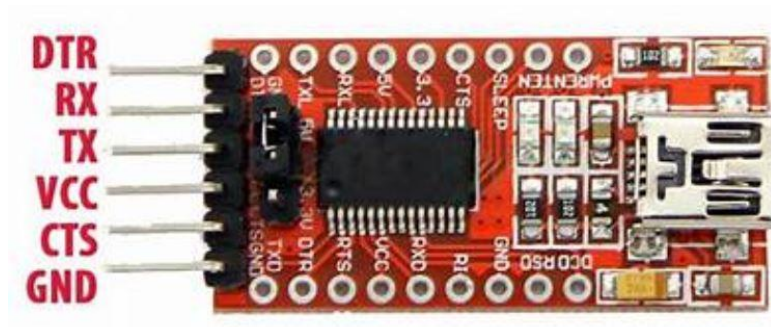


Figure 21 FTDI Adaptor

The FTDI adapter typically provides a set of pins including TX (transmit), RX (receive), GND (ground), and VCC (power), with some models also offering additional control lines like DTR (Data Terminal Ready) and CTS (Clear to Send). One of the key advantages of FTDI adapters is their reliability and ease of use—most operating systems support them with built-in or easily available drivers. They also offer selectable voltage levels (commonly 3.3V and 5V) to ensure compatibility with a wide range of devices. When using the adapter, it's important to cross-connect the TX and RX lines (i.e., connect TX from the FTDI adapter to RX on the target device, and vice versa). Due to their versatility and simplicity, FTDI adapters are widely used in electronics prototyping, firmware uploading, and serial monitoring.

5.3 AI-Powered Chatbot Web Application

We've built a standalone web application that acts like a smart assistant for our project. Using Google Gemini AI, this chatbot can answer questions in everyday language about how the robot is made and how it works. It pulls information directly from a document, so users can just type their questions — like “How does the robot move?” and get accurate answers instantly.

It's a simple and powerful way to explore everything about our spy robot project without digging through files or manuals.

System Overview

The app includes:

- A **React.js frontend** for a smooth and interactive user interface.
- A **Node.js and Express backend** that handles all the processing and communication with the AI.

Users can chat with the bot and get meaningful answers pulled directly from the document.

Key Features

PDF Reading & Chunking: The backend uses pdf-parse to extract the text from the PDF and break it into smaller, manageable pieces.

Embeddings & Search: Each piece (or "chunk") is turned into a vector using the Gemini Embedding API. When the user asks a question, it's also converted into a vector, and the system finds the most similar chunks using cosine similarity.

RAG (Retrieval-Augmented Generation): The system sends the most relevant text chunks, along with the user's question, to the Gemini AI model to generate an accurate and contextual answer.

Smart Prompting: The app chooses between a PDF-specific or general prompt depending on how closely the user's question matches the content of the document.

Technologies

Frontend:

- React.js with **Framer Motion** (for animations)
- Axios (for making API requests)
- Interactive chat UI with typing effects and timestamps



Figure 22 User Interface of web application



Figure 23 User interface with chatbot

Backend:

- Node.js and Express
- Google Gemini API (for both embeddings and generating answers)
- pdf-parse (for reading PDF content)
- Cosine similarity (to find the most relevant PDF content)
- Custom prompt templates based on query relevance



Figure 24 Response of chatbot-1

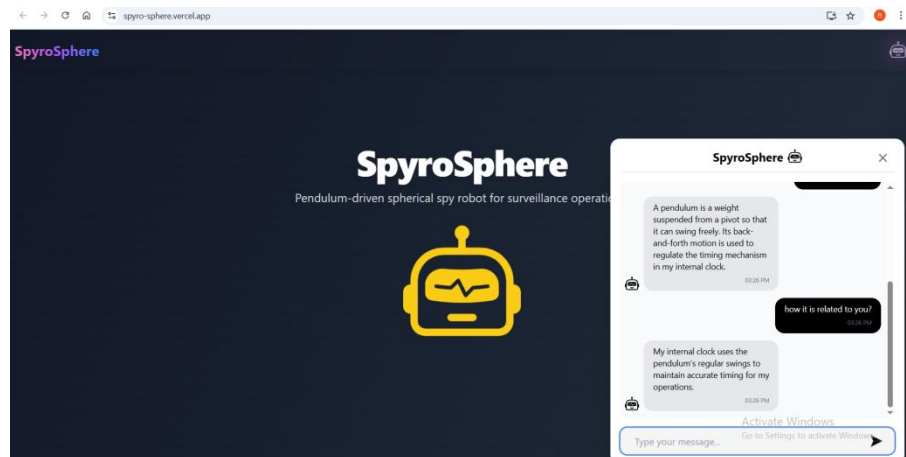


Figure 25 Response of chatbot-2

Work Flow

- When the server starts, the PDF is read and broken into smaller text chunks.
- Each chunk is embedded into a vector space using Gemini's embedding model.
- When a user types a question:
- It's also embedded into a vector.
- The system finds the most similar chunks based on cosine similarity.

- These chunks, along with the question, are sent to the Gemini model.
- The AI generates a relevant and natural-sounding response.
- The answer is displayed in the chat with a typing animation.

Benefits

- Quickly finds and explains relevant information from PDFs
- No need to manually read or search documents
- Works entirely in the browser — no installation required.

Chapter 6

INTEGRATION AND TESTING

6.1 Integrating ESP32 – CAM and L298N Motor Driver

The ESP32-CAM is connected to the L298N motor driver through several key pins that facilitate motor control. The IO12 pin of the ESP32-CAM connects to the IN1 pin of the L298N, allowing the ESP32 to send a control signal to drive the motor in one direction when activated. Similarly, the IO13 pin connects to the IN2 pin of the L298N, enabling the motor to be driven in the opposite direction, thus providing bidirectional control. Additionally, the IO2 pin of the ESP32-CAM is connected to both the ENA and ENB pins of the L298N. This connection activates the first motor channel when a high signal is sent to ENA, allowing the motor to operate, while also managing the second motor channel through ENB. Together, these connections empower the ESP32-CAM to effectively control the direction and speed of the motors connected to the L298N motor driver.

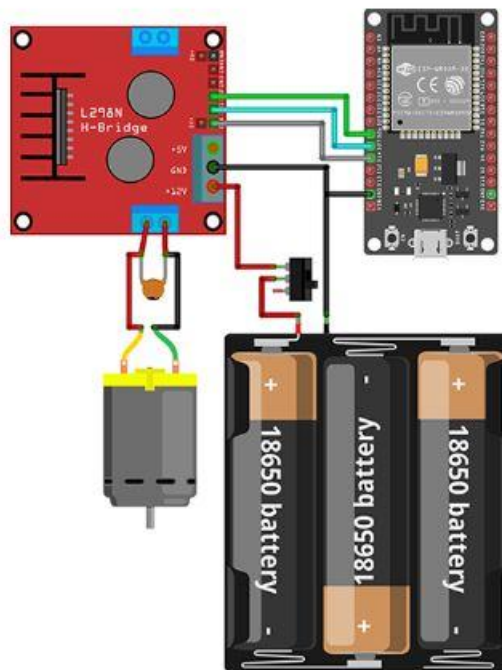


Figure 26 Interfacing ESP32 – CAM and L298N Motor Driver

6.2 Integrating ESP32-CAM and MG996R Servo Motor

To interface the ESP32-CAM with the MG996R servo motor in the given circuit, specific connections must be established to ensure proper functionality. The MG996R

servo motor has three pins: VCC, GND, and SIG. The VCC pin should be connected to a suitable power source, typically 5V, which can be sourced from the L298N motor driver or another power supply that can provide adequate current for the servo. The GND pin of the MG996R must be connected to the common ground shared with the ESP32-CAM to ensure a common reference point for the signals. The SIG pin of the MG996R is connected to the IO14 pin of the ESP32-CAM, allowing the ESP32 to send PWM signals to control the servo's position. This connection is crucial as it enables the ESP32-CAM to dictate the angle of the servo motor based on the signals it generates. By establishing these connections, the ESP32-CAM can effectively control the MG996R servo motor, facilitating various applications such as robotic movements or camera positioning.

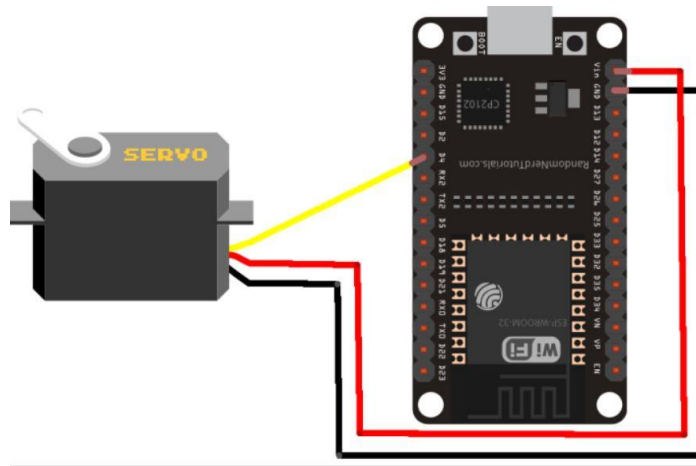


Figure 27 Interfacing ESP32-CAM and MG996R Servo Motor

6.3 Connecting ESP32-CAM with Mobile

Steps to establish a ESP32-CAM connection to mobile phone.

1. **Wi-Fi Setup:** The ESP32-CAM must be configured to connect to a Wi-Fi network. In the code, you will need to include the SSID and password of the Wi-Fi network to which the ESP32-CAM will connect. This allows the ESP32-CAM to access the internet and communicate with mobile devices.
Web Server Initialization: The ESP32-CAM can act as a web server. In the code, you will initialize a web server that listens for incoming connections. When a mobile device connects to the ESP32-CAM's IP address, it can request the video stream.
2. **Video Streaming:** The ESP32-CAM uses the MJPEG (Motion JPEG) format to stream video. In the code, you will set up the camera to capture frames and serve

them as a continuous stream over HTTP. The web server will handle requests for the video stream and send the frames to the connected client (mobile device).

3. **Accessing the Stream:** On the mobile device, you can access the live video stream by entering the IP address of the ESP32-CAM in a web browser. The ESP32-CAM will serve a webpage that displays the video stream.

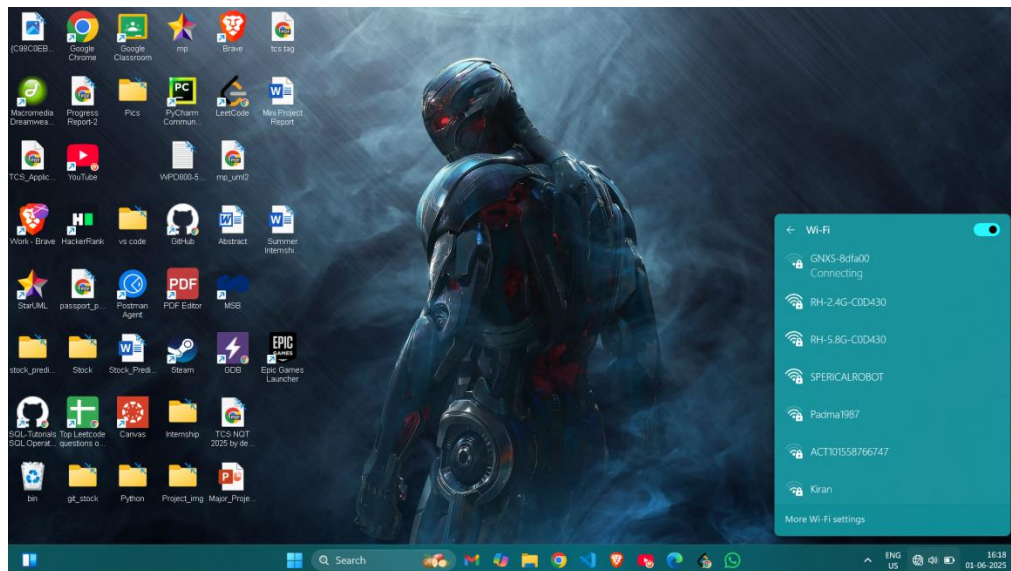


Figure 28 Wi-Fi Connection Interface



Figure 29 Connecting to Wi-Fi Network

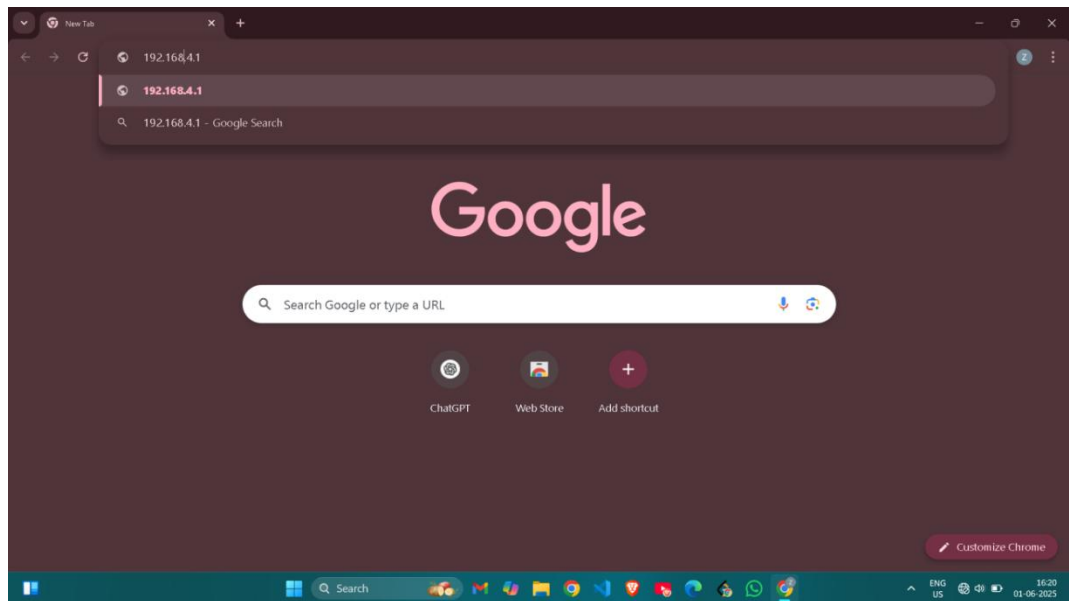


Figure 30 Entering IP Address to open web page

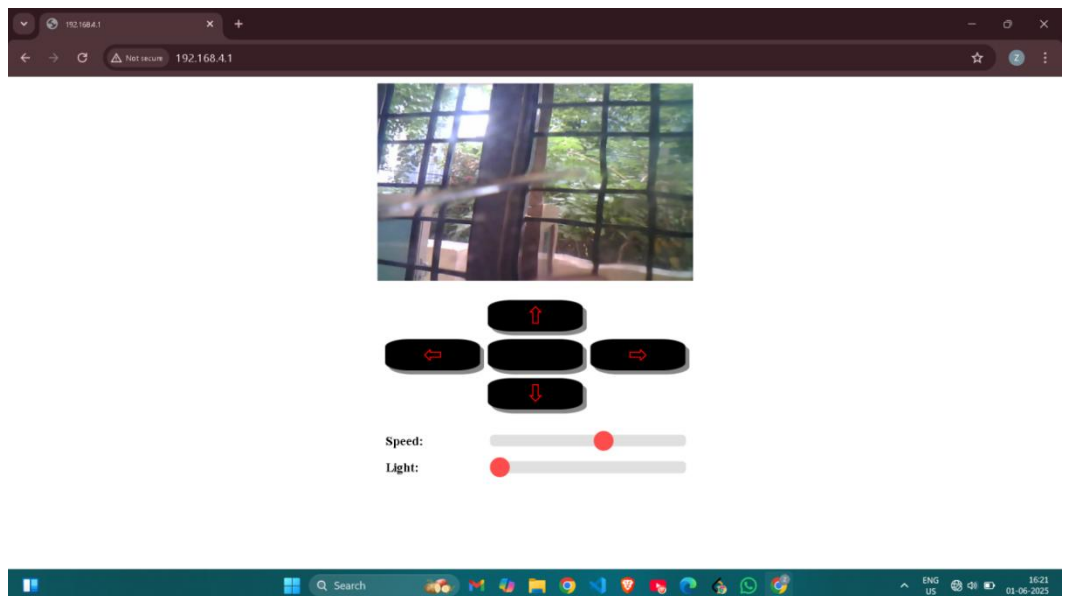


Figure 31 After Web socket Connection Web Page

Chapter 7

RESULTS

7.1 STEP-WISE OPERATIONAL WORKFLOW OF THE SURVEILLANCE

Step 1: Power ON the System

- The robot system is powered ON using the main power switch.
- This activates the on-board microcontroller (ESP32-CAM) and motor driver circuits.
- The pendulum mechanism resets to a neutral position.
- A short delay allows the ESP32-CAM to initialize and prepare the Wi-Fi connection.

Step 2: Connect to Wi-Fi

- Once powered, the ESP32-CAM either connects to a pre-configured Wi-Fi network or creates its own hotspot (based on the coding configuration).
- The user connects a smartphone, tablet, or computer to this Wi-Fi network.
- A stable connection between the user's device and the ESP32-CAM is essential for further operations.

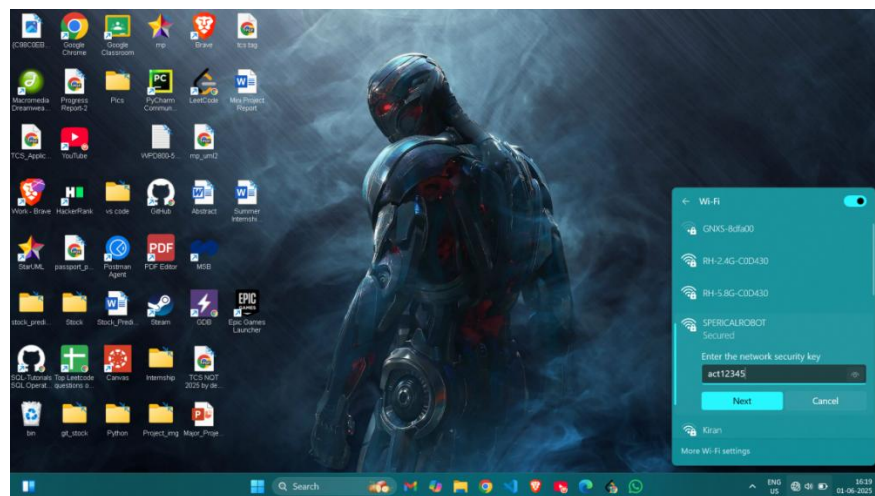


Figure 32 Wi-Fi Connection setup

Step 3: Access the Live Web Interface

- The IP address of the ESP32-CAM (usually shown in the serial monitor or pre-defined in code) is entered into a web browser.
- A custom web interface loads, displaying:
- A live video feed from the on-board camera.
- Control buttons or directional arrows for navigation.

- The video feed helps the user monitor the surroundings in real-time, making the robot effective for surveillance tasks.



Figure 33 Real time surveillance

Step 4: Send Commands to Move the Robot

- The user clicks on the direction buttons (e.g., Forward, Backward, Left, Right) on the web page.
- These commands are transmitted over Wi-Fi to the ESP32-CAM.
- The on-board microcontroller processes the input and activates motors via the motor driver, causing the internal pendulum to shift accordingly.
- The movement of the pendulum changes the robot's center of mass, making the spherical shell roll in the desired direction.
- The robot navigates over the surface while continuously streaming video, allowing the user to observe its environment.



Figure 34 Top view of Robot



Figure 35 Front view of Robot

7.2 COMPARISON TABLE

Here's the bar Table comparing the 10 research papers on spherical robots based on efficiency, image quality, and accuracy.

Ref No.	Paper Title / Authors	Image Quality	Accuracy	Efficiency	Comparison with Your Project
[1]	Jia et al. (2008)	Not focused	Moderate (actuation study)	Moderate	Our project integrates better imaging and refined actuation for precise surveillance.
[2]	Bhargavi & Manjunath (2011)	Basic camera	Low to moderate	Basic	Our robot uses higher-resolution cameras and optimized movement for better results.
[3]	MacMillan et al. (2011)	Not applicable	High (range-based nav.)	Moderate	Combines their accurate navigation with enhanced image.
[4]	Khaing & Thiha (2014)	Low	Low	Low	Our system offers superior imaging

Ref No.	Paper Title / Authors	Image Quality	Accuracy	Efficiency	Comparison with Your Project
[5]	Vashisht & Dhod	Low	Low	Basic	Our robot enhances surveillance range and real-time clarity.
[6]	Li et al. (2016)	Not applicable	High (dynamics studied)	High	Our project applies their dynamic analysis to real-time surveillance with better visuals.
[7]	Yang et al. (2020)	Not focused	High (modelling accuracy)	High	Adds practical vision systems to their mechanical precision.
[8]	Khan & Taufik (2021)	Not focused	High (3-DOF control)	Efficient	Improves by integrating surveillance capabilities into efficient control architecture.
[9]	Williams & Patel (2022)	Moderate	High	High	Our system matches their exploration ability with improved image quality.
[10]	Lee & Mishra (2023)	Not focused	High (motion planning)	High	Our robot adds vision-based feedback to motion planning.

Table 1 Comparison table of different research papers

Our project integrates the strengths of prior works—advanced actuation, accurate motion, and efficient power usage—while significantly enhancing real-time surveillance through superior imaging and smarter control system.

Chapter 8

CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The pendulum-based spherical spy robot features a simple yet efficient mechanical design that enables surveillance without relying on sensors or electronics. Its internal pendulum mechanism allows for stable motion and directional control, while the spherical outer shell ensures smooth mobility and physical protection. This makes it highly suitable for discreet operations across various terrains, offering stealth and reliability in navigation.

This purely mechanical approach is ideal for environments where electronic interference, power limitations, or risk of detection restrict traditional robotic systems. The robot's minimalistic, sensor-free design not only reduces complexity and cost but also enables easy manufacturing and maintenance. Its silent, signal-free operation enhances stealth, allowing deployment in large numbers as disposable or semi-permanent units for covert surveillance tasks.

8.2 Future scope

The pendulum-based spherical robot holds promising potential for further development and expanded applications. Future enhancements may include the integration of advanced mechanical systems such as gears, cams, or timers to enable semi-autonomous movement without the need for electronic sensors. Improvements in material selection could lead to lighter, quieter, and more durable designs, making the robot more efficient and better suited for a variety of terrains.

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APPENDIX-I

ARDUINO CODE

```
#include "esp_camera.h"
#include <Arduino.h>
#include <WiFi.h>
#include <AsyncTCP.h>
#include <ESPAsyncWebServer.h>
#include <iostream>
#include <sstream>
#include <ESP32Servo.h>

#define CENTER_PIN 14 // Replaces PAN_PIN

Servo centerServo;

struct MOTOR_PINS {
    int pinEn;
    int pinIN1;
    int pinIN2;
};

std::vector<MOTOR_PINS> motorPins = {
    {2, 12, 13}, // RIGHT_MOTOR Pins
    {2, 1, 3},  // LEFT_MOTOR Pins
};

#define LIGHT_PIN 4

#define UP 1
#define DOWN 2
#define LEFT 3
#define RIGHT 4
#define STOP 0

#define RIGHT_MOTOR 0
#define LEFT_MOTOR 1
```

```

#define FORWARD 1
#define BACKWARD -1

const int PWMFreq = 1000;
const int PWMResolution = 8;
const int PWMSpeedChannel = 2;
const int PWMLightChannel = 3;

// Camera related constants
#define PWDN_GPIO_NUM 32
#define RESET_GPIO_NUM -1
#define XCLK_GPIO_NUM 0
#define SIOD_GPIO_NUM 26
#define SIOC_GPIO_NUM 27
#define Y9_GPIO_NUM 35
#define Y8_GPIO_NUM 34
#define Y7_GPIO_NUM 39
#define Y6_GPIO_NUM 36
#define Y5_GPIO_NUM 21
#define Y4_GPIO_NUM 19
#define Y3_GPIO_NUM 18
#define Y2_GPIO_NUM 5
#define VSYNC_GPIO_NUM 25
#define HREF_GPIO_NUM 23
#define PCLK_GPIO_NUM 22

const char* ssid = "SPERICALROBOT";
const char* password = "act12345";

AsyncWebServer server(80);
AsyncWebSocket wsCamera("/Camera");
AsyncWebSocket wsCarInput("/CarInput");
uint32_t cameraClientId = 0;

const char* htmlHomePage PROGMEM = R"HTMLHOMEPAGE(

```

```

<!DOCTYPE html>
<html>
  <head>
    <meta name="viewport" content="width=device-width, initial-scale=1, maximum-
scale=1, user-scalable=no">
    ...
    <!-- (HTML content remains the same, shortened here for brevity) -->
    ...
    sendButtonInput("Pan", document.getElementById("Pan").value);
    sendButtonInput("Tilt", document.getElementById("Tilt").value);
    ...
  </body>
</html>
)HTMLHOMEPAGE";

```

```

void rotateMotor(int motorNumber, int motorDirection) {
  if (motorDirection == FORWARD) {
    digitalWrite(motorPins[motorNumber].pinIN1, HIGH);
    digitalWrite(motorPins[motorNumber].pinIN2, LOW);
  } else if (motorDirection == BACKWARD) {
    digitalWrite(motorPins[motorNumber].pinIN1, LOW);
    digitalWrite(motorPins[motorNumber].pinIN2, HIGH);
  } else {
    digitalWrite(motorPins[motorNumber].pinIN1, LOW);
    digitalWrite(motorPins[motorNumber].pinIN2, LOW);
  }
}

```

```

void moveCar(int inputValue) {
  Serial.printf("Got value as %d\n", inputValue);
  switch(inputValue) {
    case UP:
      centerServo.write(87);
      rotateMotor(RIGHT_MOTOR, BACKWARD);
      rotateMotor(LEFT_MOTOR, FORWARD);
      break;

```



```

case DOWN:
    centerServo.write(87);
    rotateMotor(RIGHT_MOTOR, FORWARD);
    rotateMotor(LEFT_MOTOR, BACKWARD);
    break;

case LEFT:
    centerServo.write(132);
    rotateMotor(RIGHT_MOTOR, BACKWARD);
    rotateMotor(LEFT_MOTOR, FORWARD);
    break;

case RIGHT:
    centerServo.write(42);
    rotateMotor(RIGHT_MOTOR, BACKWARD);
    rotateMotor(LEFT_MOTOR, FORWARD);
    break;

case STOP:
default:
    centerServo.write(87);
    rotateMotor(RIGHT_MOTOR, STOP);
    rotateMotor(LEFT_MOTOR, STOP);
    break;
}
}

void handleRoot(AsyncWebServerRequest *request) {
    request->send_P(200, "text/html", htmlHomePage);
}

void handleNotFound(AsyncWebServerRequest *request) {
    request->send(404, "text/plain", "File Not Found");
}

```

```

void onCarInputWebSocketEvent(AsyncWebSocket *server,
AsyncWebSocketClient *client, AwsEventType type, void *arg, uint8_t *data, size_t
len) {
    switch (type) {
        case WS_EVT_CONNECT:
            Serial.printf("WebSocket client #%u connected from %s\n", client->id(), client-
>remoteIP().toString().c_str());
            break;
        case WS_EVT_DISCONNECT:
            Serial.printf("WebSocket client #%u disconnected\n", client->id());
            moveCar(0);
            ledcWrite(PWMLightChannel, 0);
            centerServo.write(90);
            break;
        case WS_EVT_DATA: {
            AwsFrameInfo info = (AwsFrameInfo)arg;
            if (info->final && info->index == 0 && info->len == len && info->opcode ==
WS_TEXT) {
                std::string myData((char *)data, len);
                std::istringstream ss(myData);
                std::string key, value;
                std::getline(ss, key, ',');
                std::getline(ss, value, ',');
                Serial.printf("Key [%s] Value[%s]\n", key.c_str(), value.c_str());
                int valueInt = atoi(value.c_str());
                if (key == "MoveCar") {
                    moveCar(valueInt);
                } else if (key == "Speed") {
                    ledcWrite(PWMSpeedChannel, valueInt);
                } else if (key == "Light") {
                    ledcWrite(PWMLightChannel, valueInt);
                } else if (key == "Pan") {
                    centerServo.write(valueInt);
                }
            }
            break;
        }
    }
}

```

```

    case WS_EVT_PONG:
    case WS_EVT_ERROR:
    default:
        break;
    }
}

void onCameraWebSocketEvent(AsyncWebSocket *server, AsyncWebSocketClient
*client, AwsEventType type, void *arg, uint8_t *data, size_t len) {
    switch (type) {
        case WS_EVT_CONNECT:
            Serial.printf("WebSocket client #%u connected from %s\n", client->id(), client-
>remoteIP().toString().c_str());
            cameraClientId = client->id();
            break;
        case WS_EVT_DISCONNECT:
            Serial.printf("WebSocket client #%u disconnected\n", client->id());
            cameraClientId = 0;
            break;
        case WS_EVT_DATA:
        case WS_EVT_PONG:
        case WS_EVT_ERROR:
        default:
            break;
    }
}

void setupCamera() {
    camera_config_t config;
    config.ledc_channel = LEDC_CHANNEL_4;
    config.ledc_timer = LEDC_TIMER_2;
    config.pin_d0 = Y2_GPIO_NUM;
    config.pin_d1 = Y3_GPIO_NUM;
    config.pin_d2 = Y4_GPIO_NUM;
    config.pin_d3 = Y5_GPIO_NUM;
    config.pin_d4 = Y6_GPIO_NUM;
    config.pin_d5 = Y7_GPIO_NUM;

```

```

config.pin_d6 = Y8_GPIO_NUM;
config.pin_d7 = Y9_GPIO_NUM;
config.pin_xclk = XCLK_GPIO_NUM;
config.pin_pclk = PCLK_GPIO_NUM;
config.pin_vsync = VSYNC_GPIO_NUM;
config.pin_href = HREF_GPIO_NUM;
config.pin_sscb_sda = SIOD_GPIO_NUM;
config.pin_sscb_scl = SIOC_GPIO_NUM;
config.pin_pwdn = PWDN_GPIO_NUM;
config.pin_reset = RESET_GPIO_NUM;
config.xclk_freq_hz = 20000000;
config.pixel_format = PIXFORMAT_JPEG;
config.frame_size = FRAMESIZE_VGA;
config.jpeg_quality = 10;
config.fb_count = 1;

esp_err_t err = esp_camera_init(&config);
if (err != ESP_OK) {
    Serial.printf("Camera init failed with error 0x%x", err);
    return;
}

if (psramFound()) {
    heap_caps_malloc_extmem_enable(20000);
    Serial.printf("PSRAM initialized.\n");
}
}

void sendCameraPicture() {
    if (cameraClientId == 0) return;

    camera_fb_t * fb = esp_camera_fb_get();
    if (!fb) {
        Serial.println("Frame buffer could not be acquired");
        return;
    }
}

```

```

wsCamera.binary(cameraClientId, fb->buf, fb->len);
esp_camera_fb_return(fb);

while (true) {
    AsyncWebsocketClient * clientPointer = wsCamera.client(cameraClientId);
    if (!clientPointer || !(clientPointer->queueIsFull())) break;
    delay(1);
}
}

void setUpPinModes() {
    centerServo.attach(CENTER_PIN);

    ledcSetup(PWMSpeedChannel, PWMFreq, PWMResolution);
    ledcSetup(PWMLightChannel, PWMFreq, PWMResolution);

    for (int i = 0; i < motorPins.size(); i++) {
        pinMode(motorPins[i].pinEn, OUTPUT);
        pinMode(motorPins[i].pinIN1, OUTPUT);
        pinMode(motorPins[i].pinIN2, OUTPUT);
        ledcAttachPin(motorPins[i].pinEn, PWMSpeedChannel);
    }

    moveCar(STOP);

    pinMode(LIGHT_PIN, OUTPUT);
    ledcAttachPin(LIGHT_PIN, PWMLightChannel);
}

void setup(void) {
    setUpPinModes();
    WiFi.softAP(ssid, password);
    IPAddress IP = WiFi.softAPIP();
    Serial.print("AP IP address: ");
    Serial.println(IP);
}

```

```

server.on("/", HTTP_GET, handleRoot);
server.onNotFound(handleNotFound);

wsCamera.onEvent(onCameraWebSocketEvent);
server.addHandler(&wsCamera);

wsCarInput.onEvent(onCarInputWebSocketEvent);
server.addHandler(&wsCarInput);

server.begin();
Serial.println("HTTP server started");

setupCamera();
}

void loop() {
  wsCamera.cleanupClients();
  wsCarInput.cleanupClients();
  sendCameraPicture();
  Serial.printf("SPIRam   Total   heap   %d,   SPIRam   Free   Heap   %d\n",
ESP.getPsramSize(), ESP.getFreePsram());
}

```